3.1: Scientific terms for solar energy and PV systems

3.1.1: Guide to Solar Energy and related Concepts

Irradiance | Insolation | Solar Constant | Direct Normal Irradiance | Beam Radiation | Diffused Horizontal Irradiance | Albedo Radiation

Global Horizontal Irradiance | Irradiance at Tilted Surface | Air Mass | Latitude Angle | Solar Hour Angle | Declination Angle

Equinox | Solstice | Azimuth Angle | Zenith Angle

Irradiance

- Irradiance (or power density) is defined as the solar energy received by Earth's surface per unit area.
- It's unit of measurement is: (W/m^2)

Insolation

- Insolation or irradiation (or energy density) is defined as the amount of solar energy received by Earth's surface over a given duration of time.
- It's unit of measurement is: (Wh/m^2)

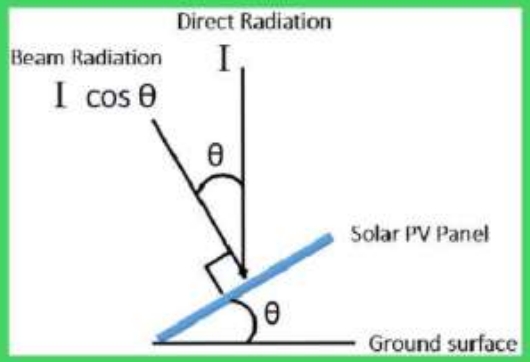
Solar Constant

- Solar constant is the amount of energy that normally falls on a unit area ($1m^2$) of the Earth's atmosphere per second when the Earth is at its mean distance from the Sun.
- The value of the Solar constant is 1367 W/m^2.
- It's unit of measurement is: (W/m^2)

Direct Normal Irradiance

- Direct normal irradiance is the solar radiation reaching Earth's surface without being absorbed or scattered.
- It's unit of measurement is: (W/m^2)

Beam Radiation



- Beam radiation is defined as the Cosine component of the Direct Normal Irradiance.
- It's unit of measurement is: (W/m^2)

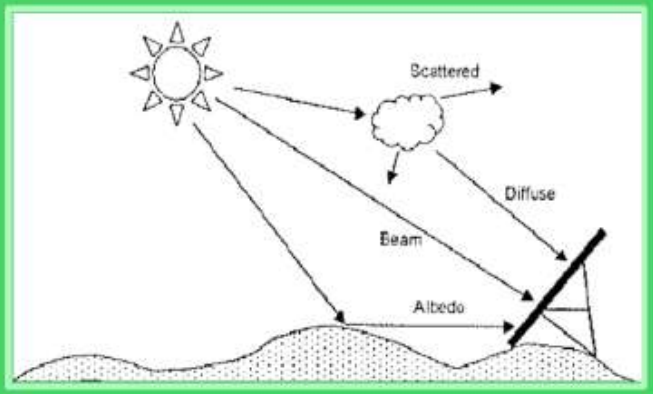
Diffused Horizontal Irradiance

- Diffused horizontal irradiance is defined as the sum of all scattered radiations.
- It's unit of measurement is: (W/m^2)

Albedo Radiation

- Albedo radiation is defined as the radiation (part of diffused and direct radiation) that gets reflected by Earth and other objects on the Earth.
- It's unit of measurement is: (W/m^2)

Global Horizontal Irradiance

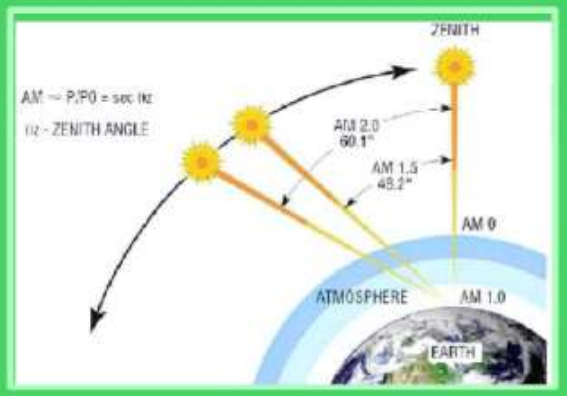


- Global horizontal irradiance is defined as the sum of the diffused radiation, direct radiation and albedo radiation.
- It's unit of measurement is: (W/m^2)

Irradiance at Tilted Surface

- Irradiance at tilted surface is defined as the radiation falling on any tilted surface.
- It's unit of measurement is: (W/m^2)

Air Mass



- Air Mass is the distance travelled by solar radiation in Earth's atmosphere.

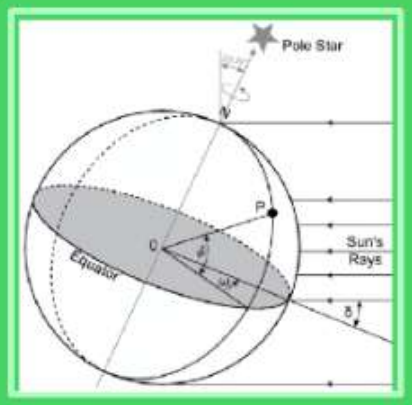
Latitude Angle

- Latitude angle (or angle of latitude) is defined as the angle drawn between the lines joining the centre of Earth to the site with its projection on the equatorial plane. For India, it is considered to be positive.
- It is denoted by 'φ'.

Solar Hour Angle

- Solar hour angle is defined as the angular measurement of time.
- Conventionally, it is taken positive in morning and negative in the afternoon.
- It is denoted by 'ω'.

Declination Angle



- Declination angle is defined as the angle drawn between the lines joining the centre of Earth to the centre of Sun having its projection on the equatorial plane of the Earth.
- It is denoted by 'δ'.

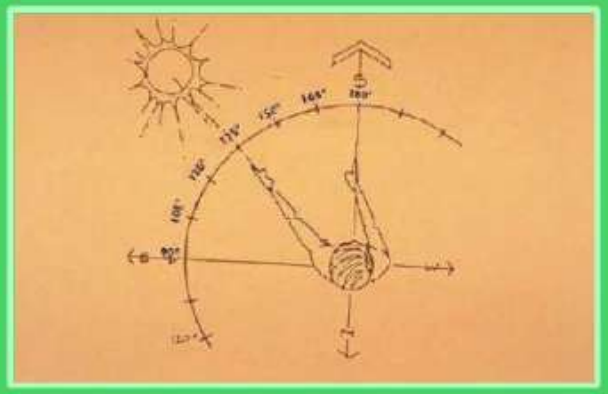
Equinox

- Equinox refers to "equal night".
- It is a day when the number of hours of daylight equals the number of hours of night.
- The vernal equinox, usually March 21, signals the onset of Spring, while the autumnal equinox, usually September 21, signals the onset of Autumn.

Solstice

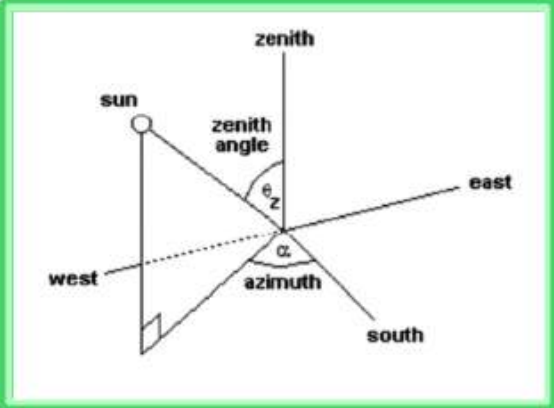
Solstice is a day when the sun is at the highest point in the sky (summer solstice, 21 June) or at the lowest point in the sky (winter solstice, 22 December).

Azimuth Angle



- Azimuth angle is defined as the angle between the sun's rays and true South.
- A positive solar azimuth angle indicates a position East of South, and a negative azimuth angle indicates West of South.

Zenith Angle



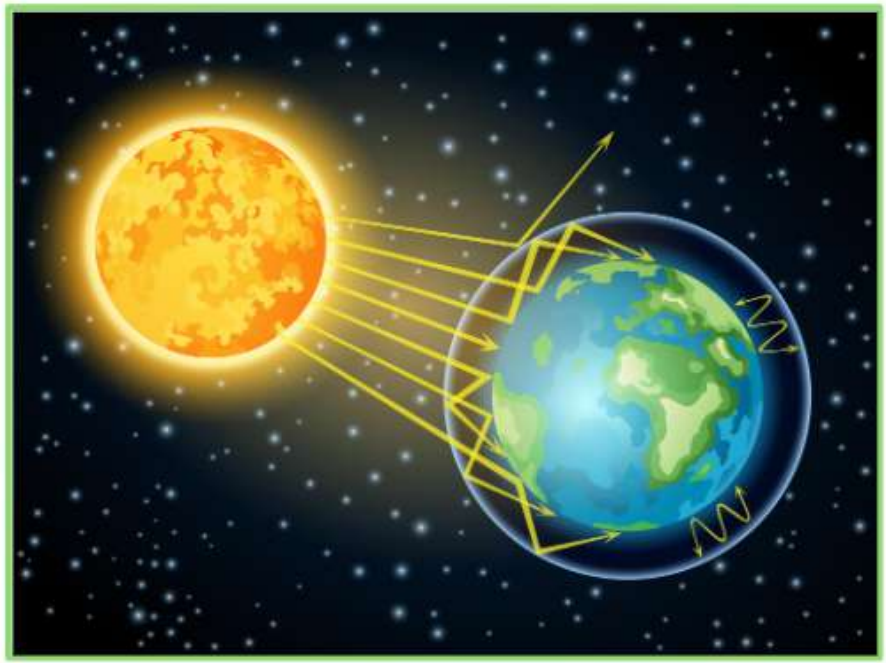
- Zenith angle is defined as the angle drawn between the Sun and the vertical.

3.2: Sun path diagram and solar radiation

3.2.1: Solar Radiation and Sun Path Diagram

- Solar Radiation is the energy emitted by the sun, consisting of visible light, infrared, and ultraviolet rays.

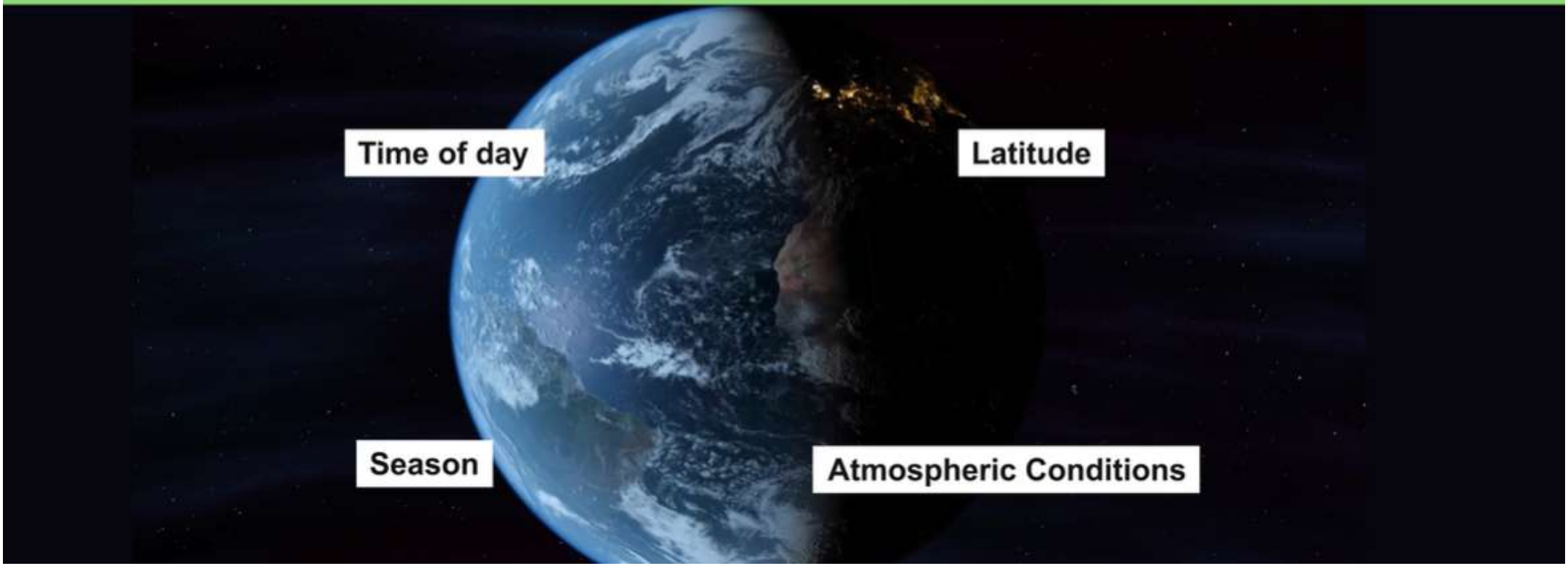
- Sun Path Diagram provides a visual representation of the Sun's path across the sky at a specific location over a certain period.



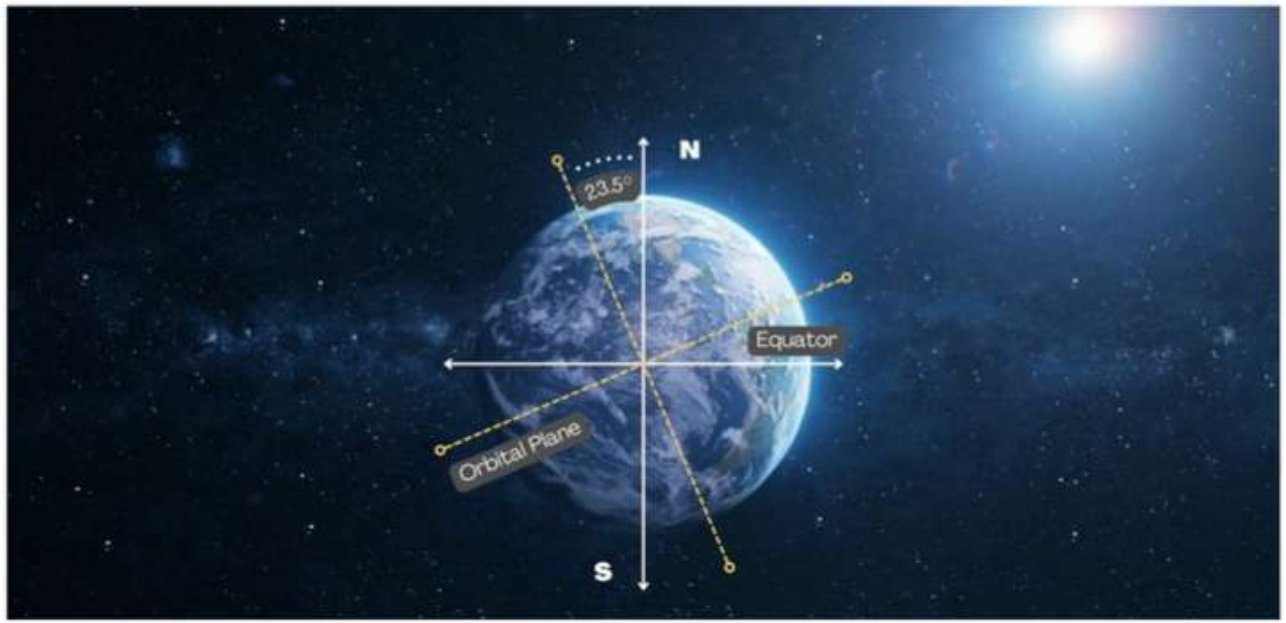
What is Solar Radiation?



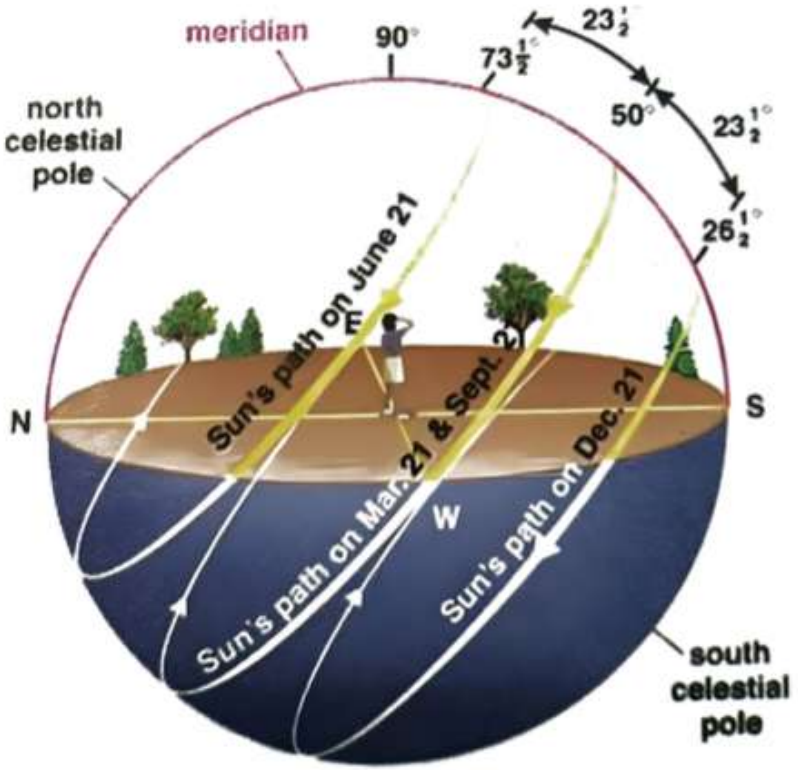
Factors Affecting Solar Radiation



What is Sun Path Diagram?

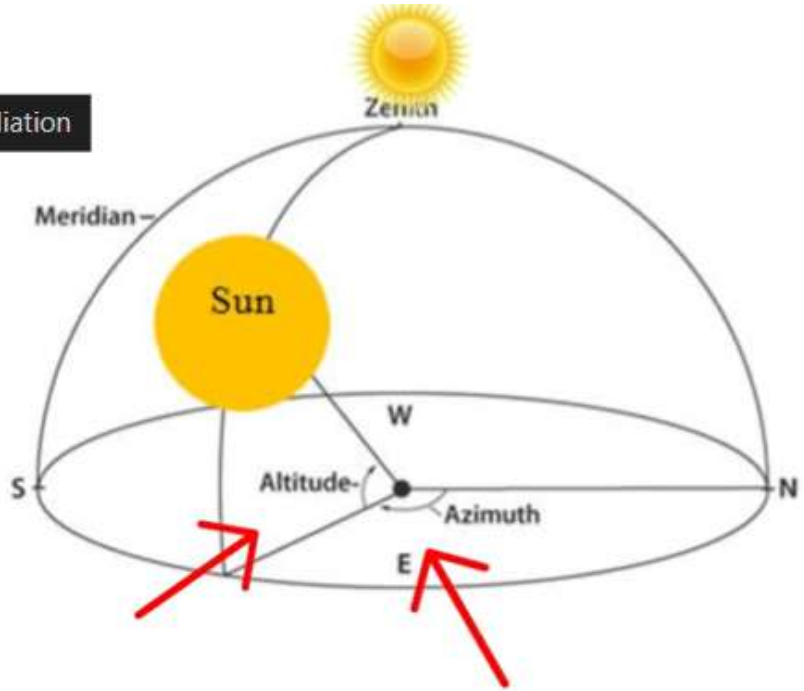


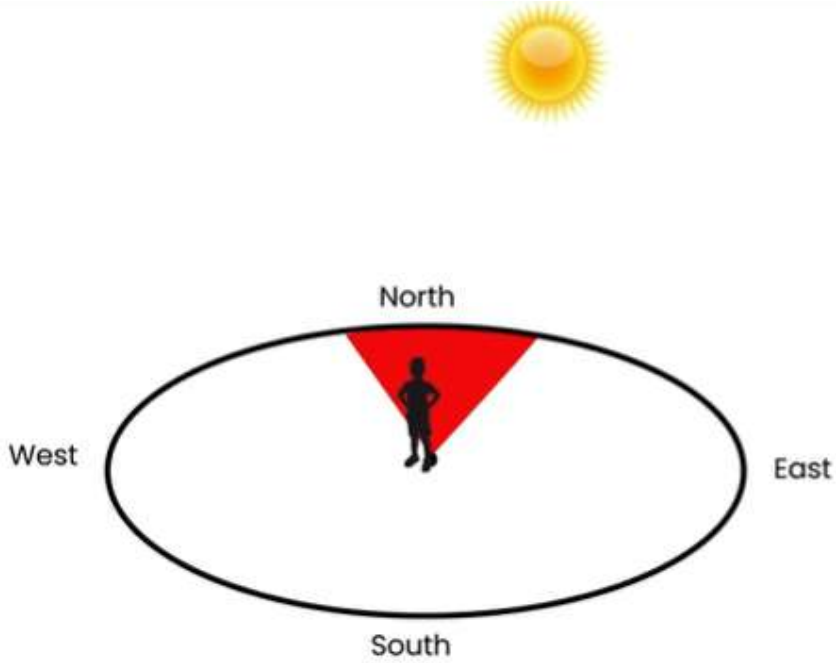
A Sun Path Diagram is a graphical representation of the Sun's path across the sky – at a specific location and specific time.



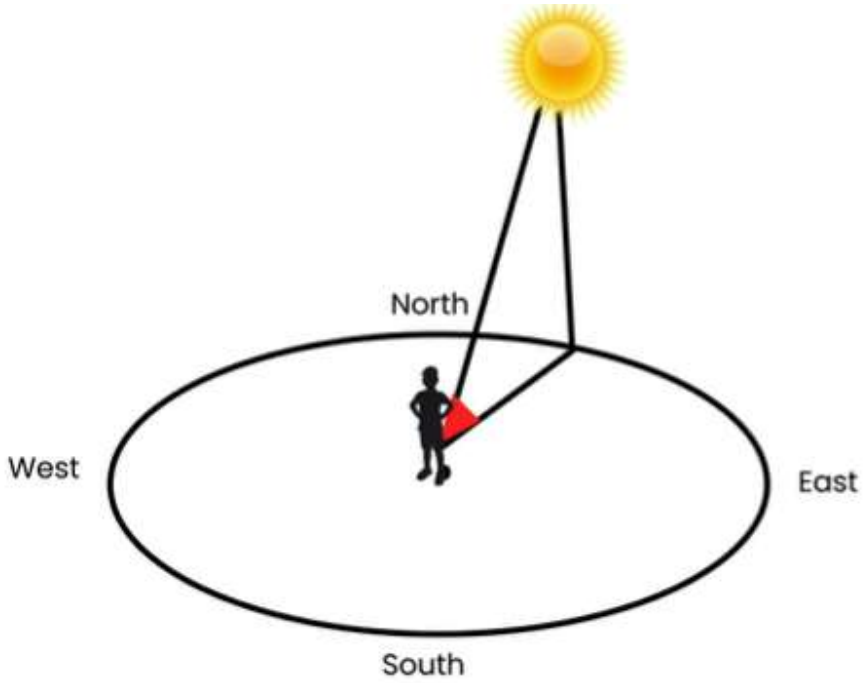
The Sun Path as seen here from a person's point of view is different for **June 21** and **December 21**. But path is same for **March 21** and **September 21**.

n Diagram and Solar Radiation

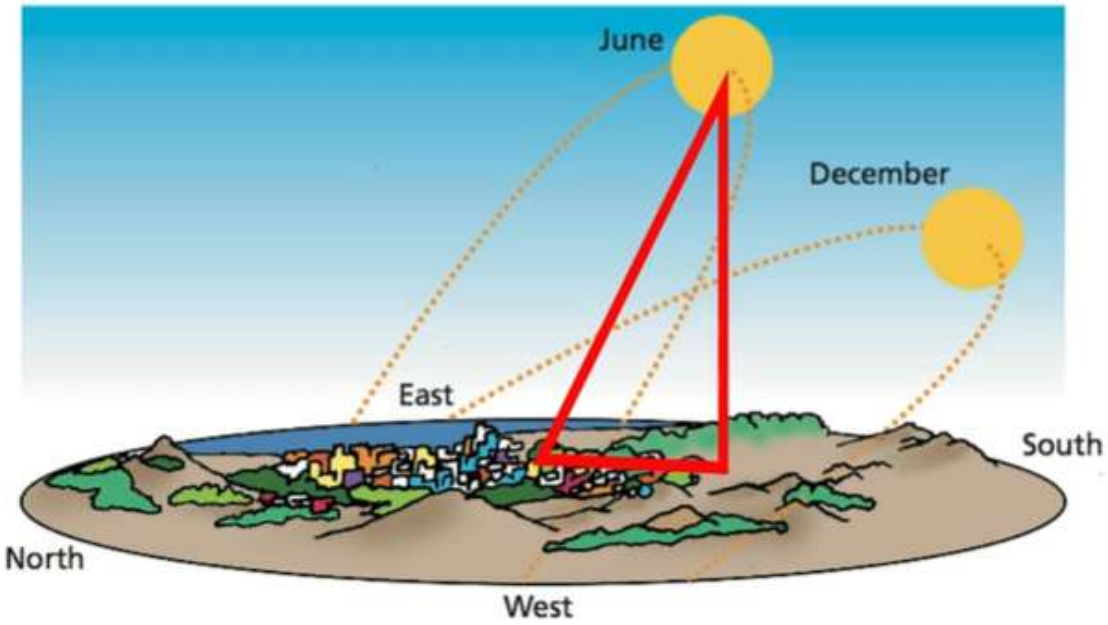


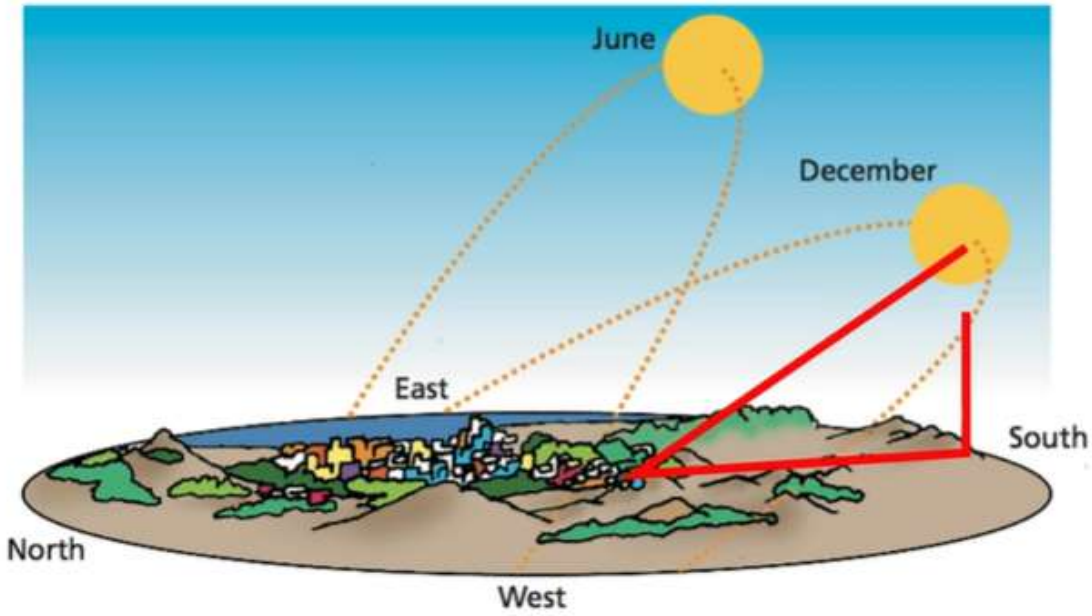


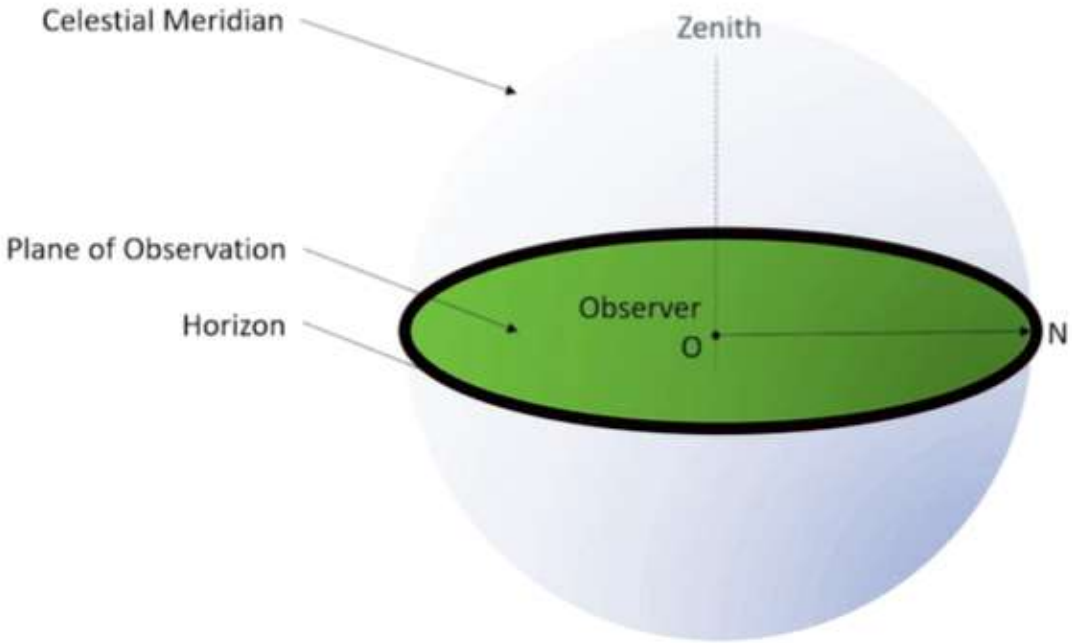
Azimuth Angle - angle between the sun's rays and true South.

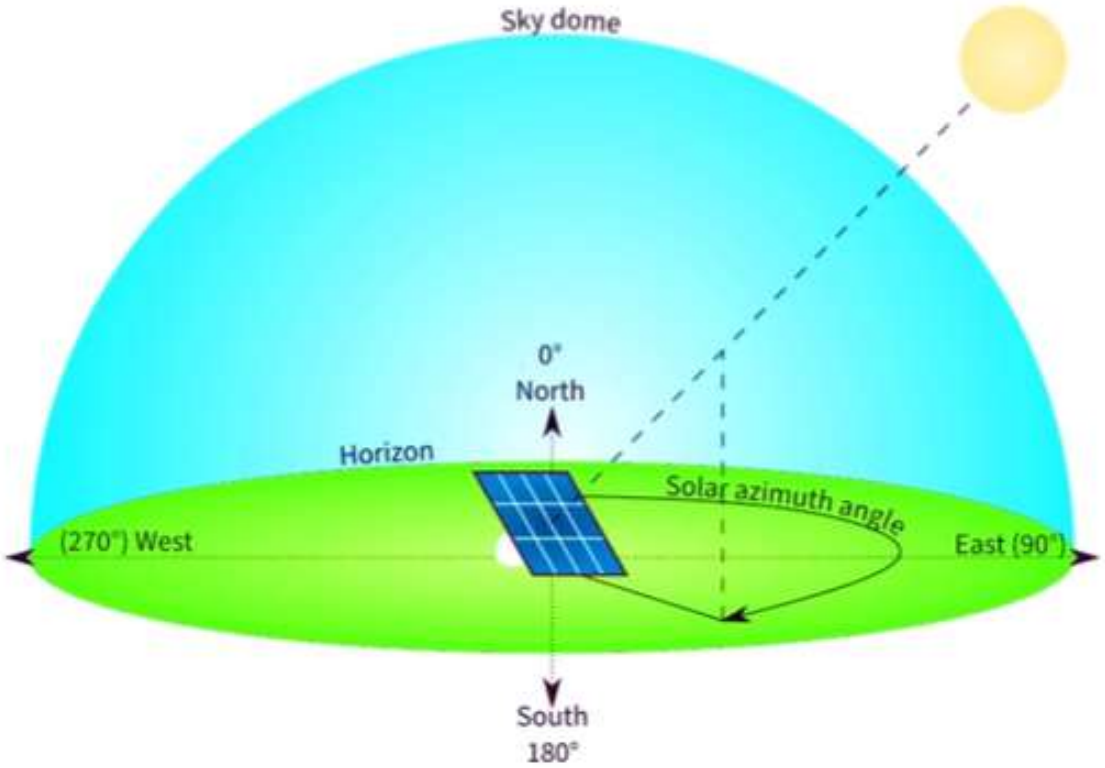


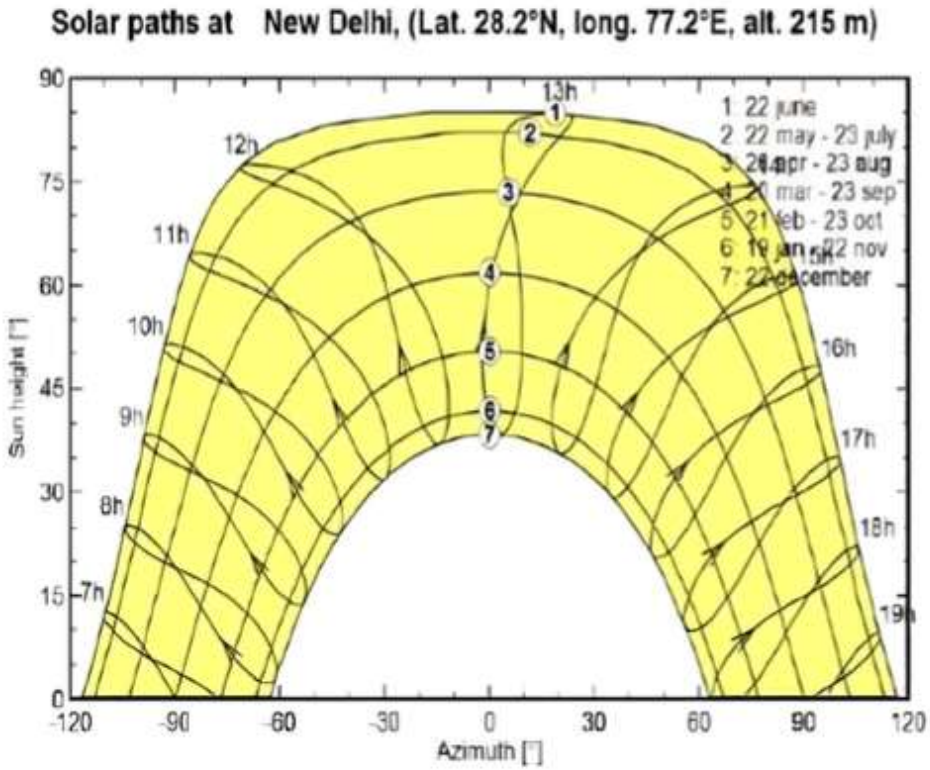
Sun Height or Altitude – angular distance between sun and horizon.


June
December
East
South
North
West

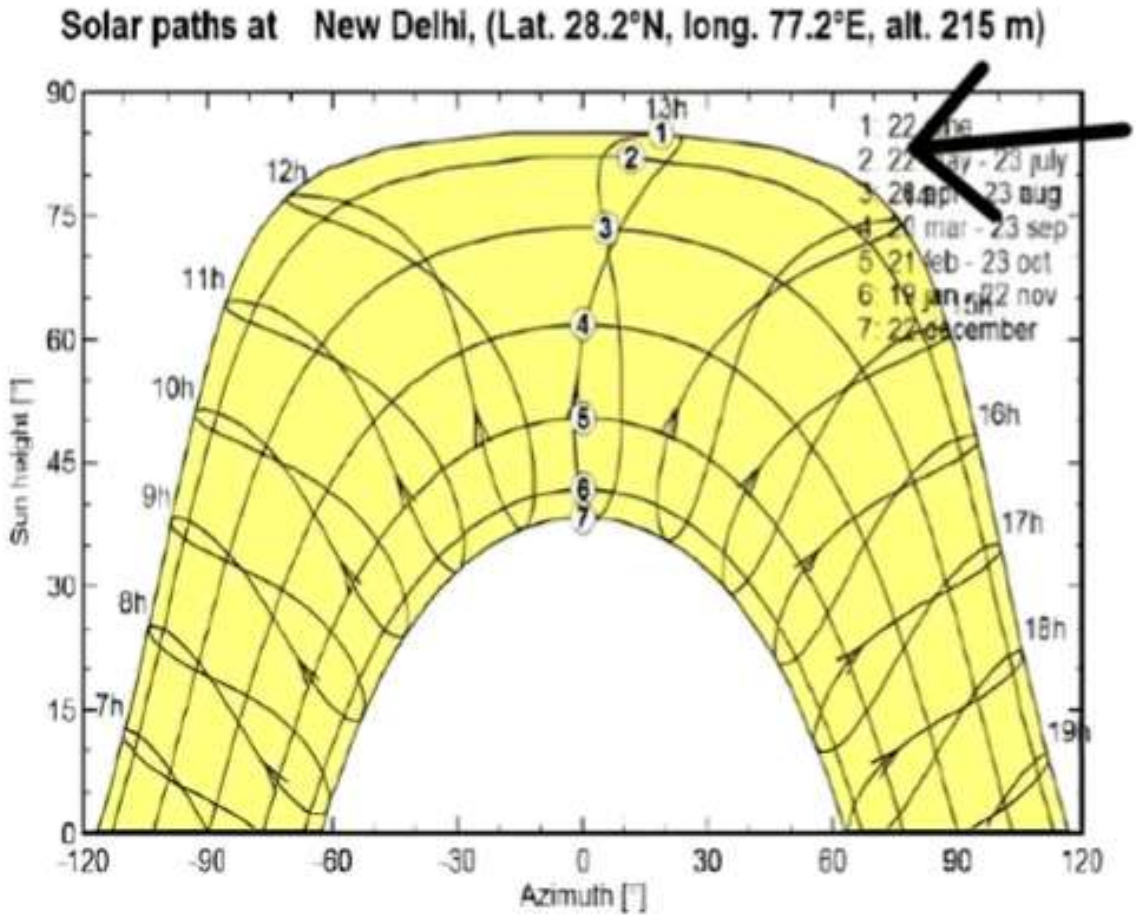

June
December
East
South
North
West


Celestial Meridian
Zenith
Plane of Observation
Horizon
Observer
O
N


Sky dome
0°
North
Horizon
Solar azimuth angle
(270°) West
East (90°)
South
180°



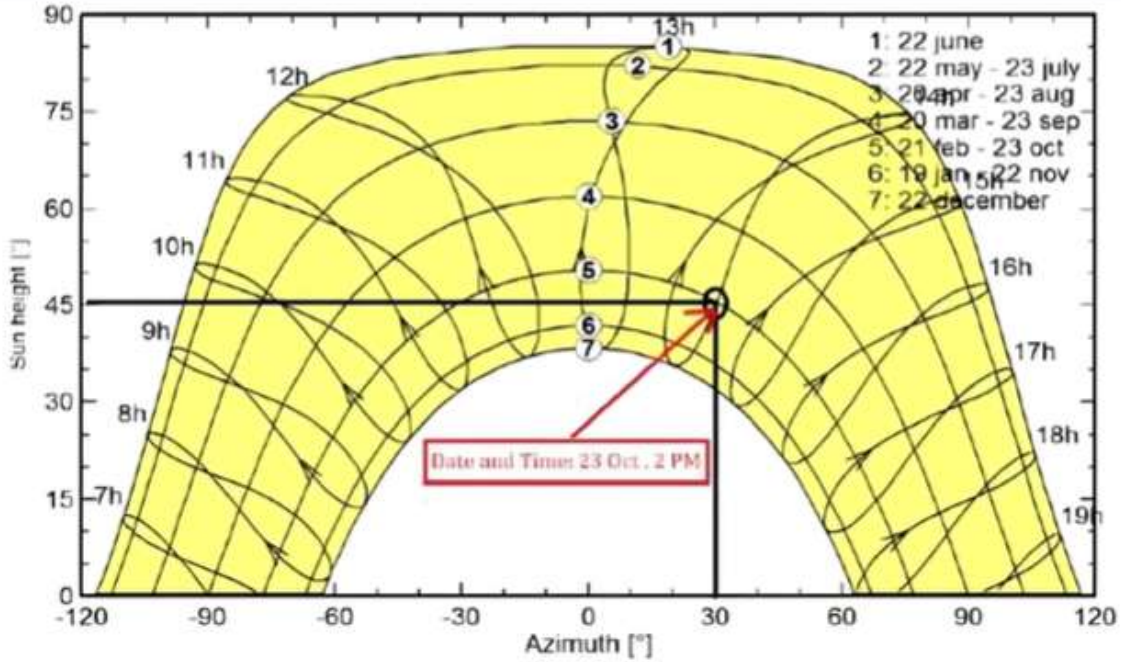
22 June
22 May- 23 July
20 April – 23 August
20 March – 23 September
21 February – 23 October
19 January – 22 November
22 December



Summer solstice

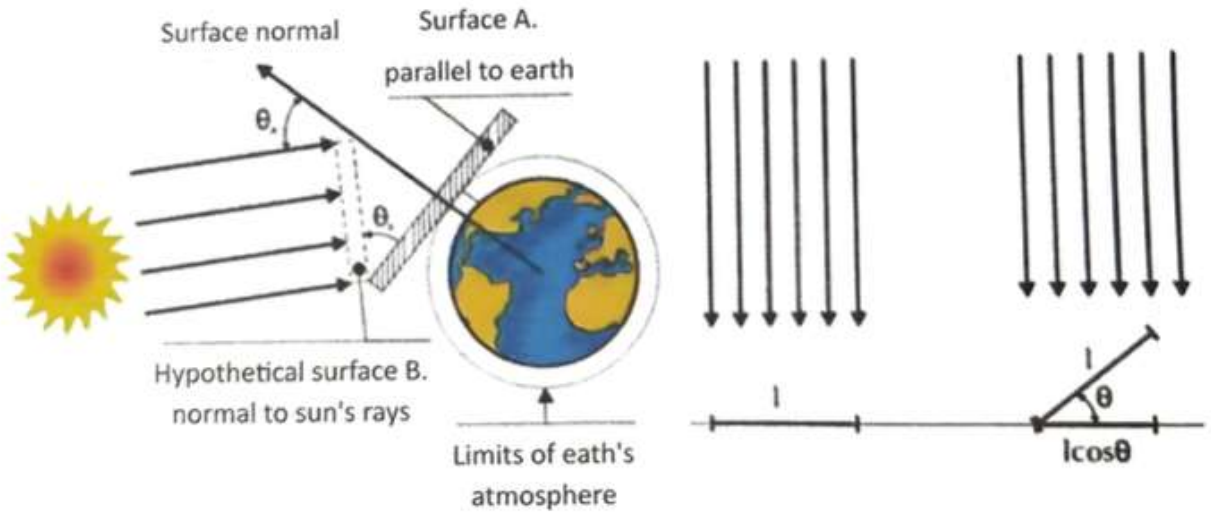
June 22

How to interpret Sun Path Diagrams?



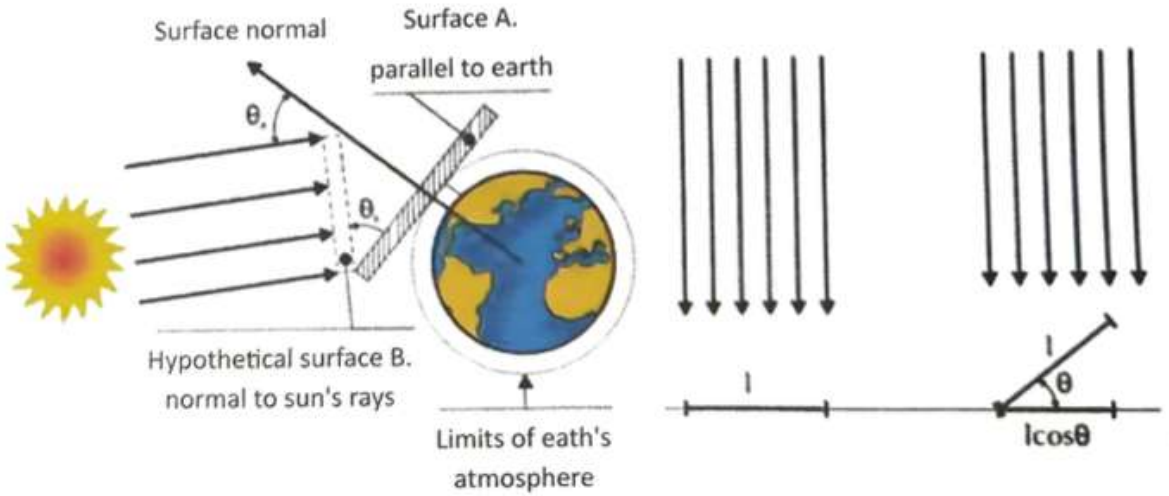
Conclusion : by interpreting these Sun path diagrams, you can determine the Sun's position throughout the day and year.

Tilt Angle and Cosine Effect

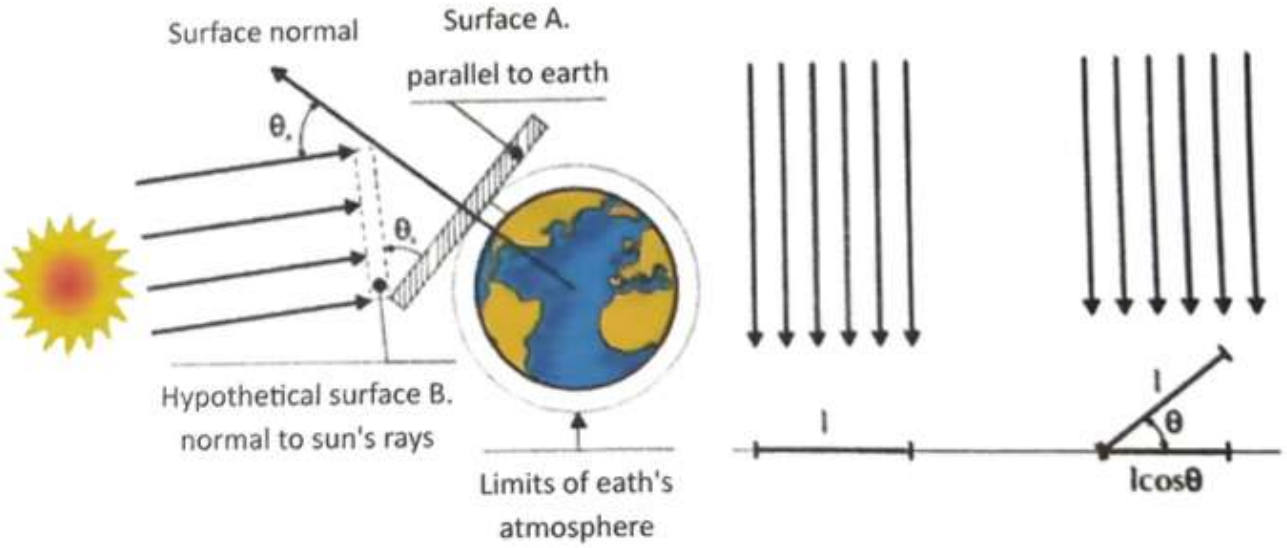


As a Solar PV installer – it is important to know the tilt angle which allows you to optimize solar panel orientation for maximum energy capture and thus conversion to electricity.

Tilt Angle and Cosine Effect

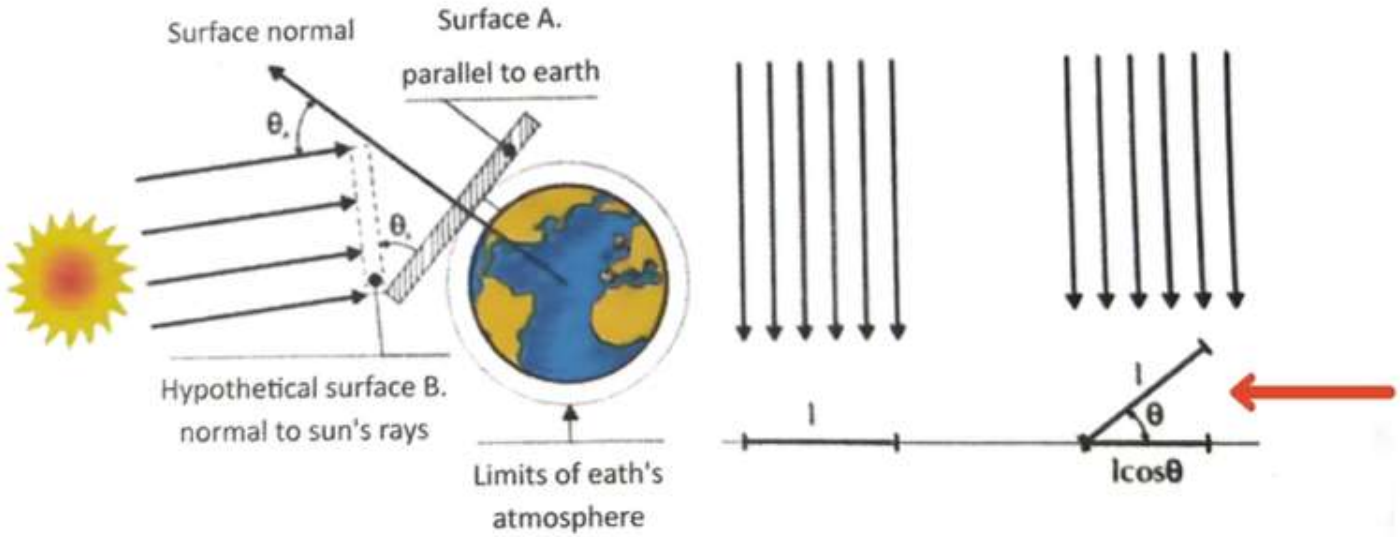


The tilt angle for solar panels refers to the angle at which the panels are installed relative to the horizontal plane.

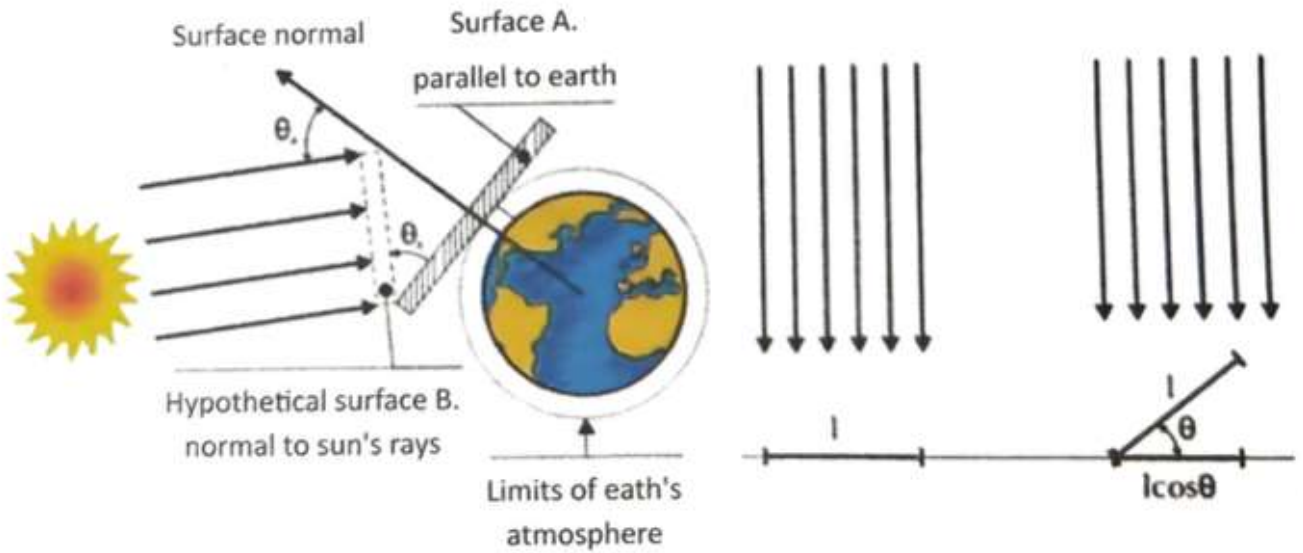


This is optimized based on factors such as:

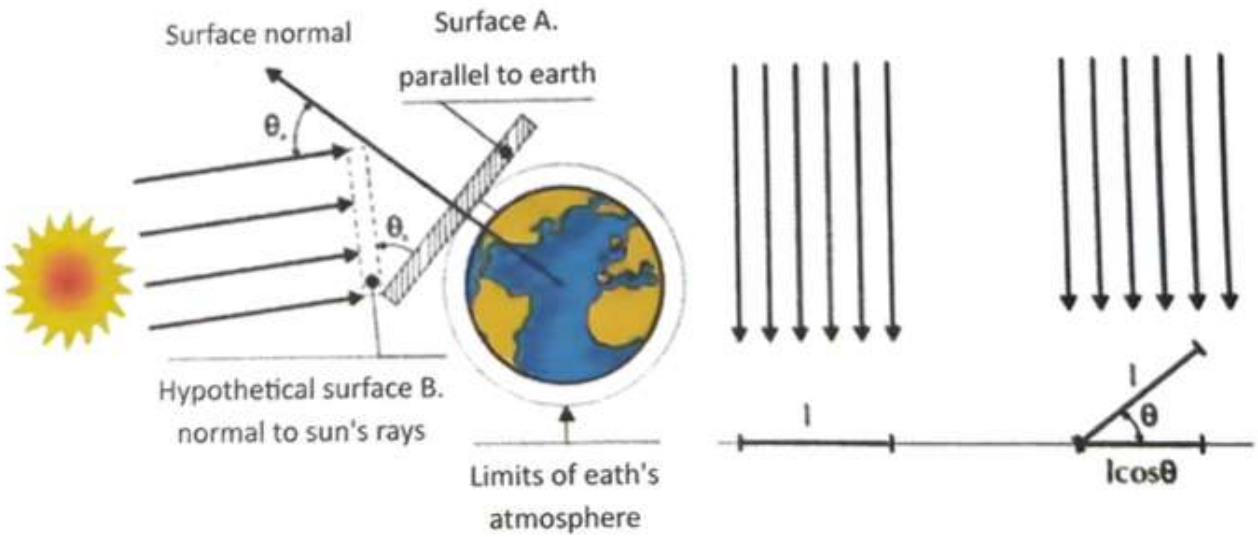
- Geographical location
- Season
- Desired energy output.



We can calculate the tilt angle using the cosine effect.

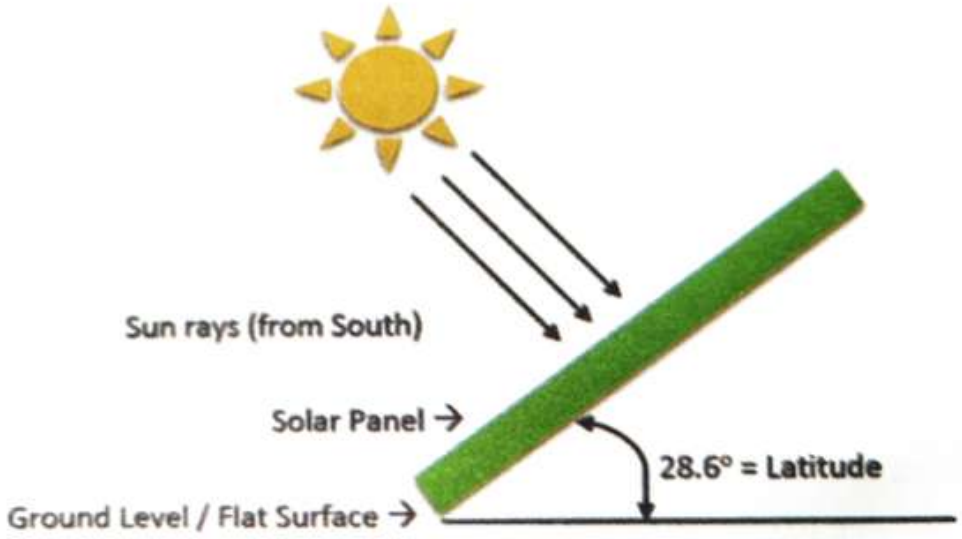


The principle of cosine effect explains how the angle of incidence of sunlight affects the intensity of solar radiation received by a surface.



The Cosine Effect states that the intensity of solar radiation on a surface is directly proportional to the cosine of the angle of incidence between the sunlight and the surface normal.

Solar Panel Performance With Different Orientations and Tilt Angles

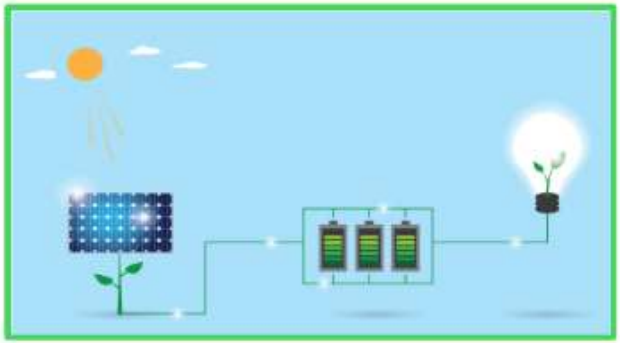


For Example, India lies between the Latitude of 80 degrees North to 360 degrees North and Delhi's latitude is 28.6 degrees North. So, the tilt angle of the solar panel will be 28.6 degrees.

3.3: Components of a solar PV system

3.3.1: Solar PV System and their Components

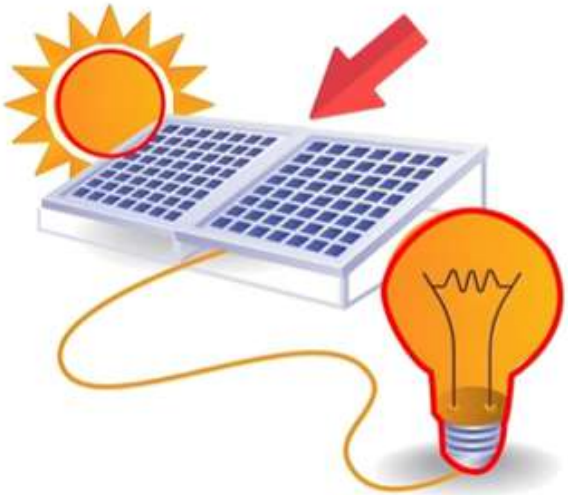
- A solar PV system comprises several essential components working together to convert sunlight into usable electricity.

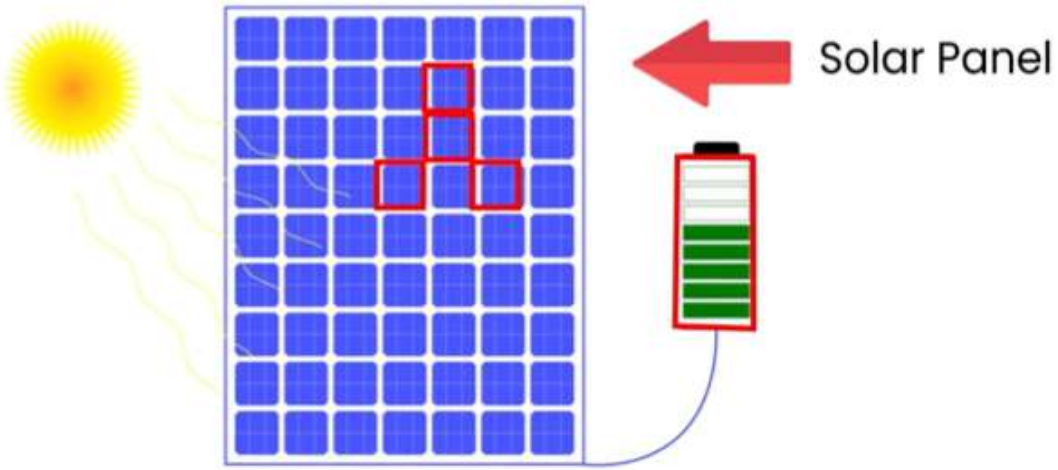


- Each component plays a crucial role in harnessing solar energy efficiently, making solar PV systems a sustainable and renewable energy solution for various applications.



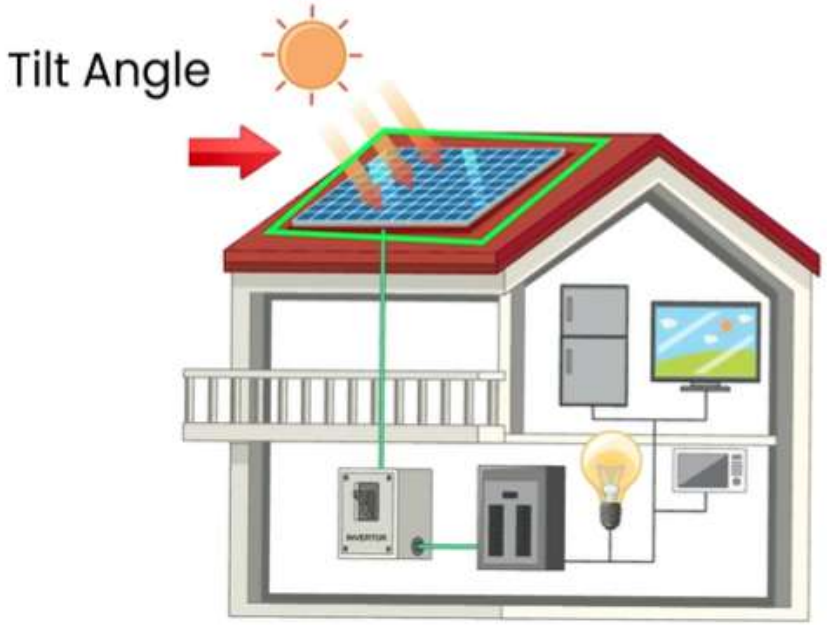
PhotoVoltaic Panel





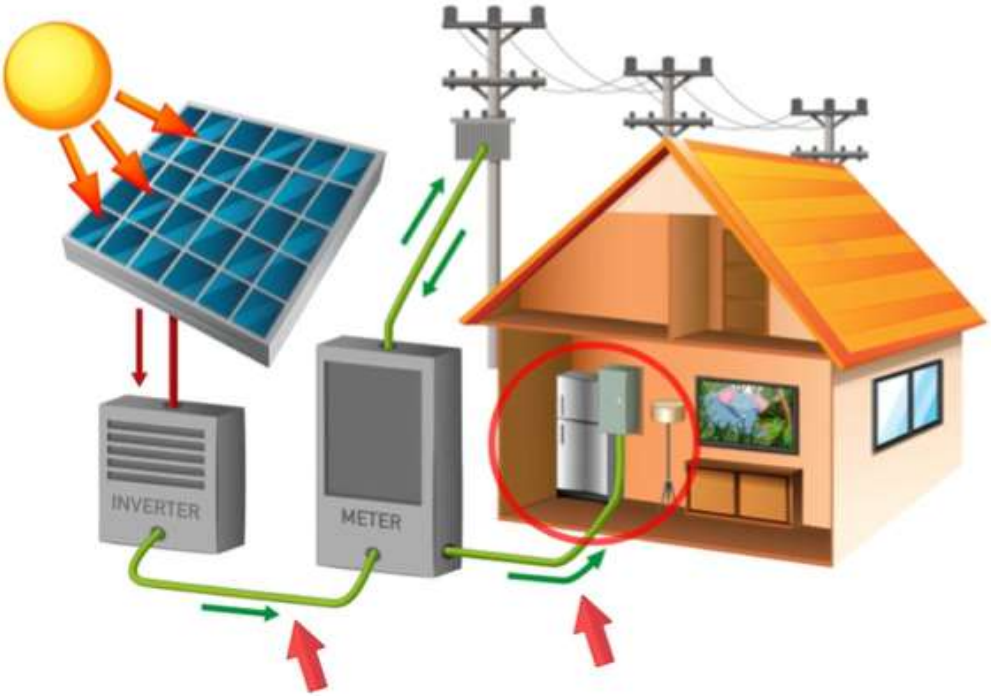
A Solar PV System works by converting sunlight into electricity through Photovoltaic panels. These panels are made up of individual solar cells that generate direct current (DC) electricity when exposed to sunlight.

Photovoltaic Panel



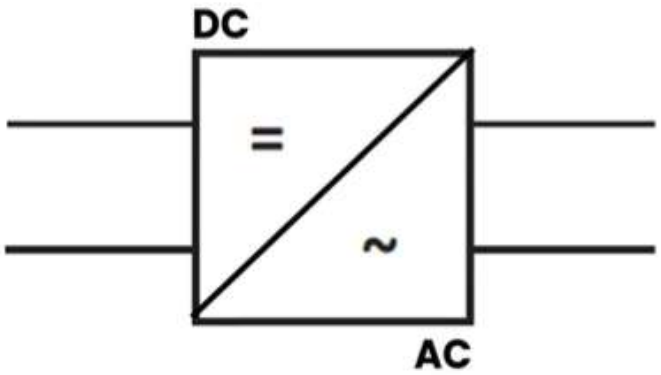
The Solar PV Panels are typically installed on rooftops or open spaces where they can receive maximum sunlight exposure throughout the day. Proper orientation and tilt angle are crucial for optimizing energy generation.

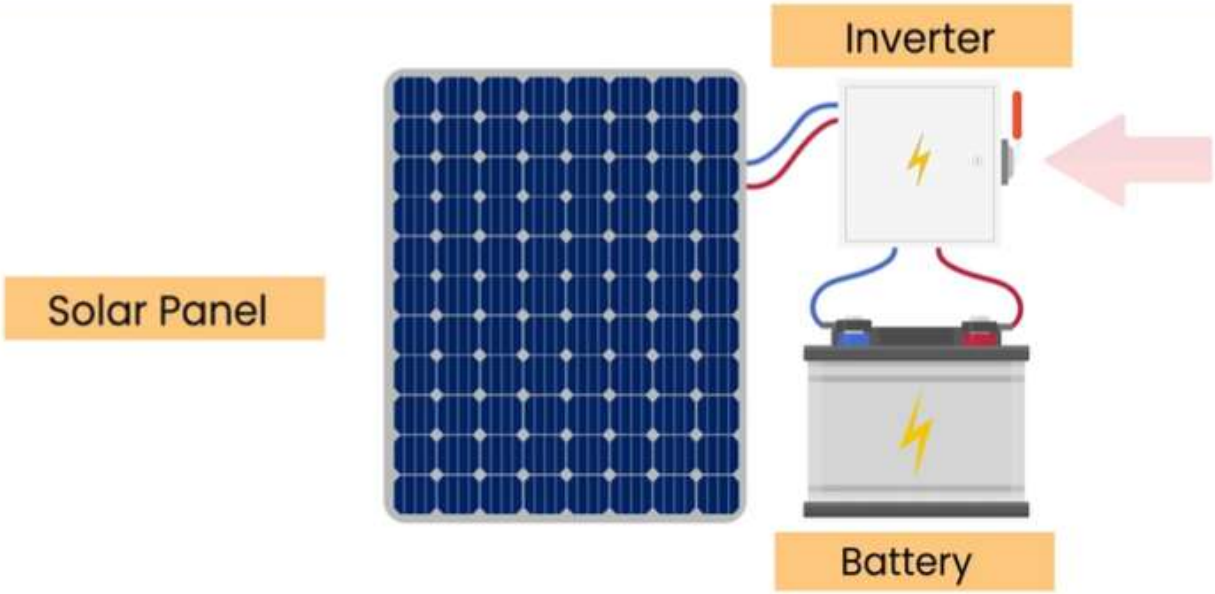
Inverter



- A solar inverter is a device that converts direct current or DC electricity, which is what a solar panel generates,to alternating current or AC electricity,which is what the main electrical power grid uses.
- The conversion happens as per the required frequency and voltage level of the house's electrical system.
- It is placed between the PV panel and AC grid and connects them through cables.

Inverter

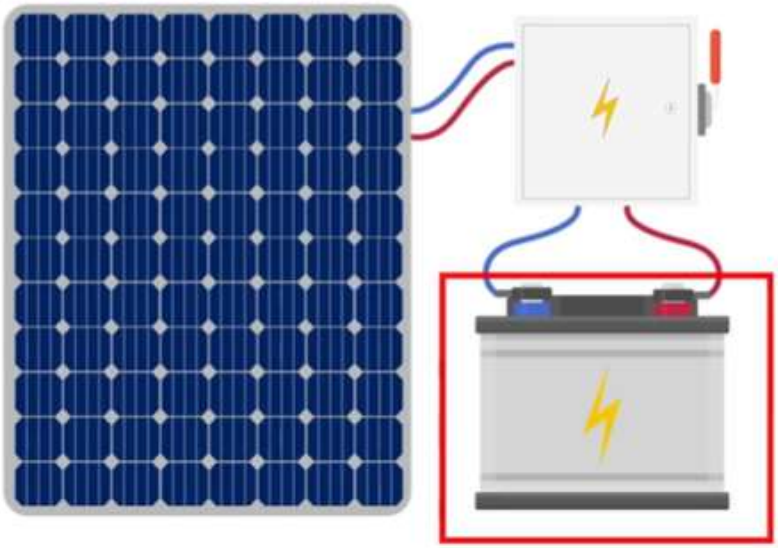




Besides conversion from DC to AC, an Inverter also functions for adjustment of operating point to maximum power point through an MPP tracker, and also ensures grid monitoring and management.

Battery

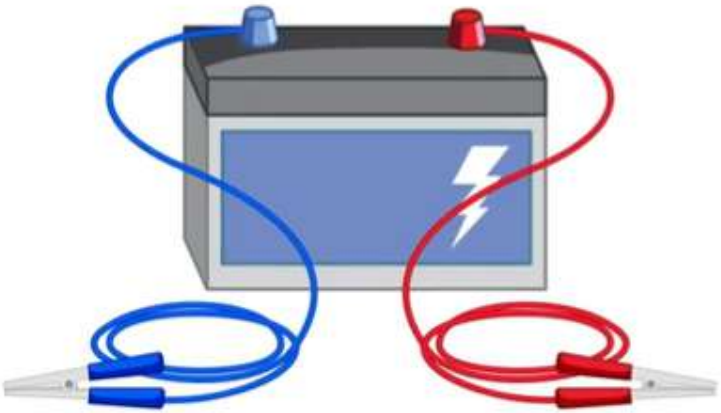
Solar Panel



Inverter

Battery

Batteries are essential components of off-grid or hybrid solar PV systems. They store excess electricity generated during the day for use during periods of low sunlight or at night.

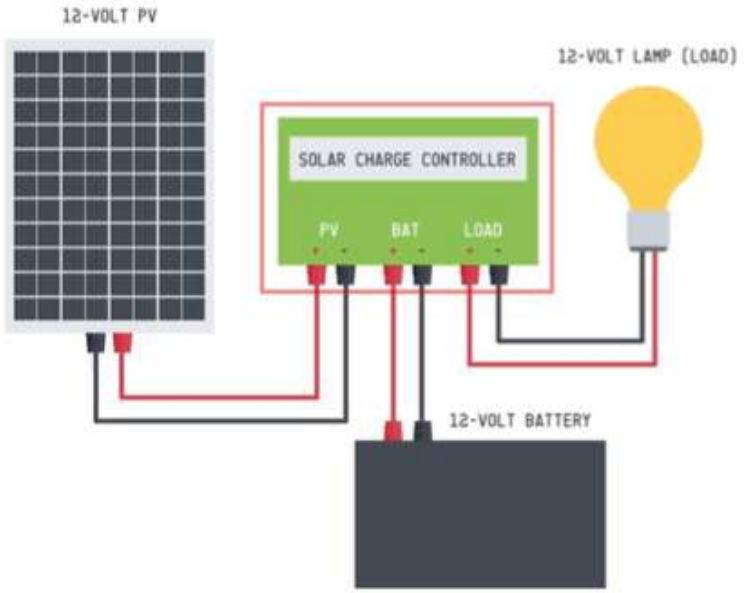


There are various types of batteries available :

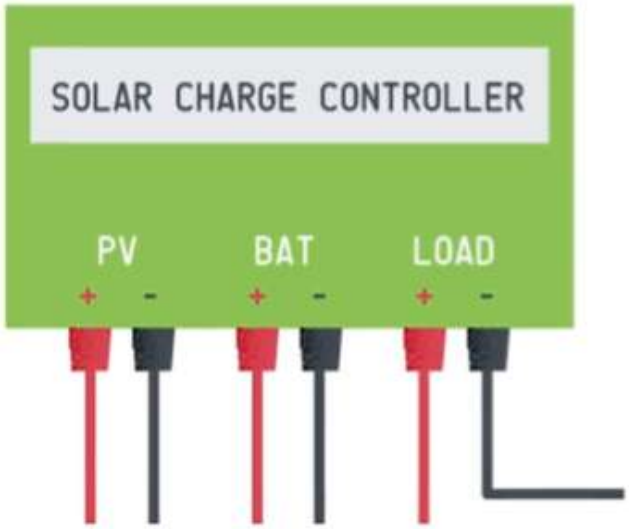
- Nickel Cadmium (NiCd)
- Nickel Metal Hydride (NiMH)
- Lead Acid Batteries
- Lithium-Ion (Li ion)
- Lithium-Ion Polymer

The most common type of battery in standalone systems are Lead acid batteries – most cost effective, handle large & small charging currents.

Charge Controller

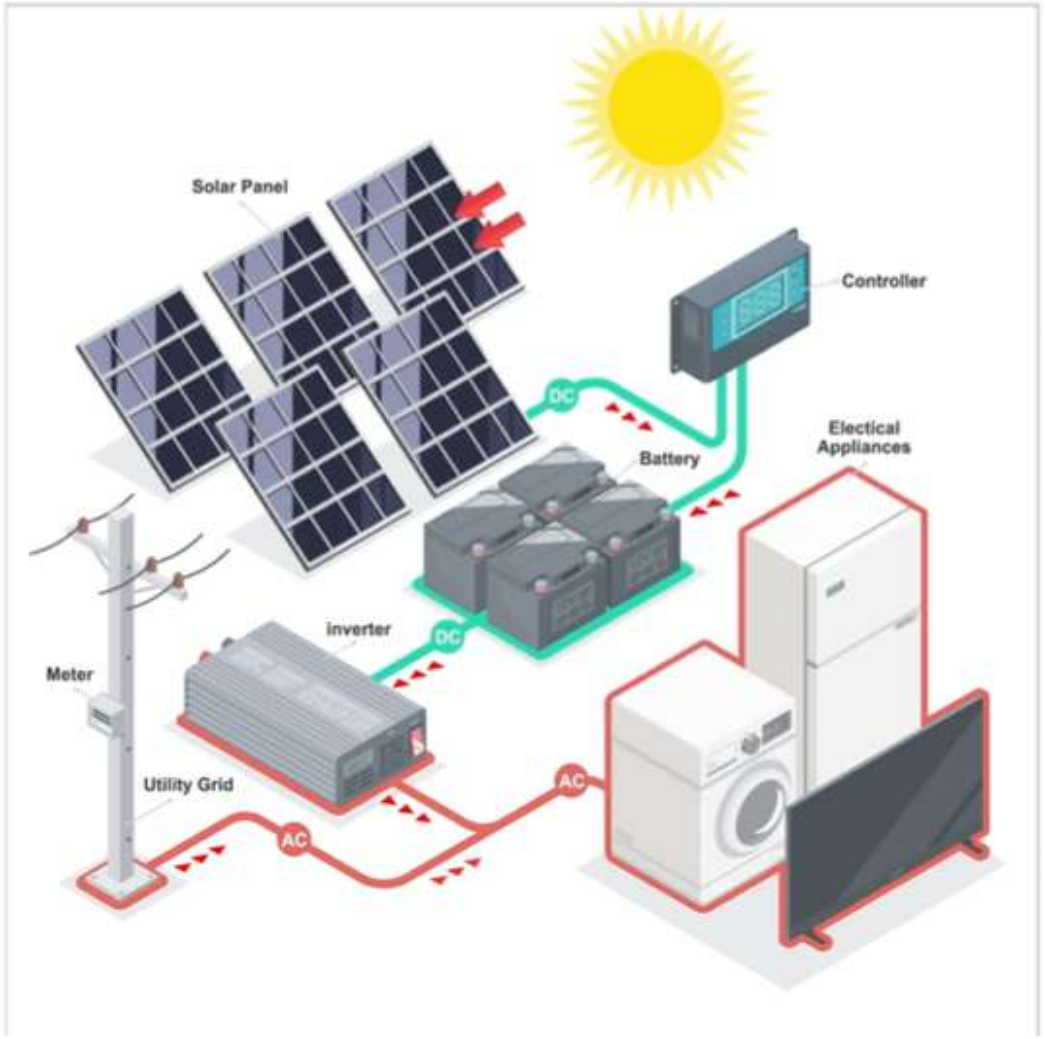


Charge Controllers regulate the charging and discharging of batteries, ensuring their longevity and optimal performance.



Additional essential functions of Charge Controllers

- Overcharging Protection
- Deep Discharge Protection
- Preventing Unintentional Discharge
- Optimal Charging of Batteries
- Reverse Polarity Protection
- Battery Indication
- Short Circuit Protection

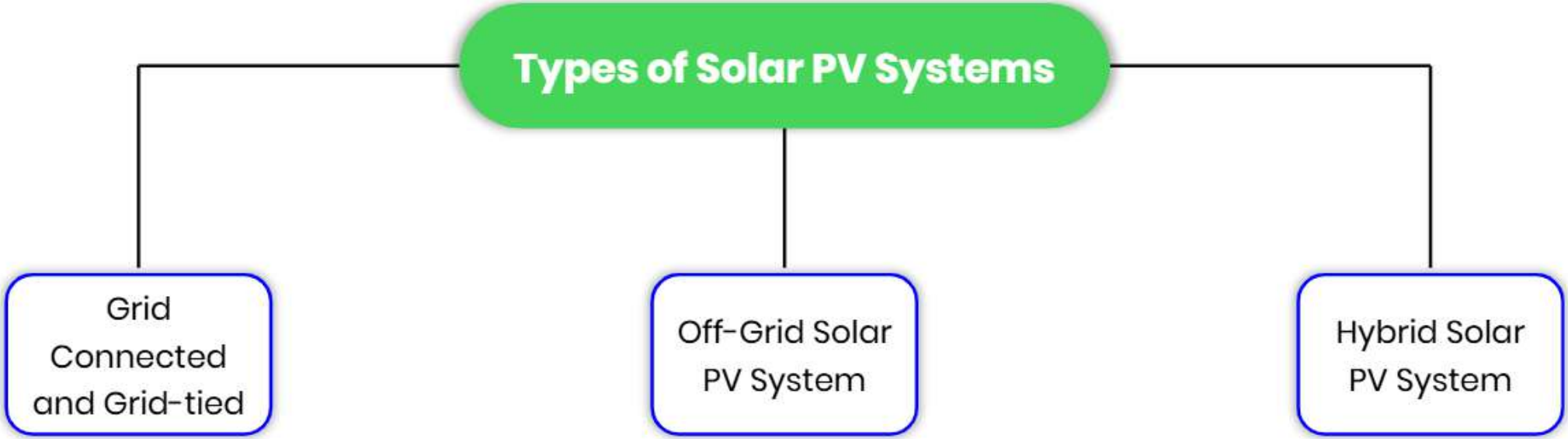

Solar Panel
Controller
DC
Battery
Electical
Appliances
inverter
DC
Meter
Utility Grid
AC
AC



A solar PV system consists of solar panels, inverters, batteries, charge controllers, and electrical wiring.

3.4: Types of solar PV system

3.4.1: Types of Solar PV Systems



Each type caters to different energy needs and environmental conditions, providing sustainable solutions for diverse applications.

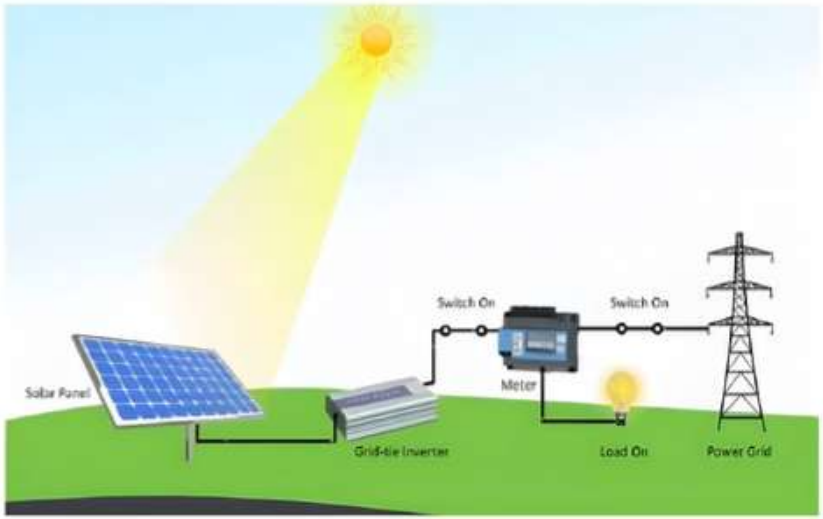
Types of Solar PV Systems

- Grid Connected and Grid-Tied
- Off-Grid
- Hybrid



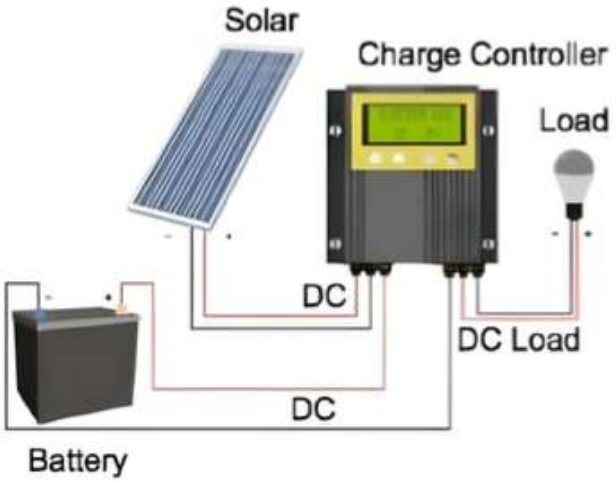
Grid-Connected and Grid-Tied Systems

- Connected to the power grid (utility grid) - main source of electricity supply.
- Generate electricity from sunlight and feed it into the grid, typically offsetting the user's electricity consumption.
- Allow excess electricity to be sold back to the grid, in exchange of credit or compensation.
- Most common type - ideal for residential, commercial, and industrial applications.
- **"Grid-Tied"** - systems without storage.
- **"Grid-Connected"** – systems with energy storage capabilities like battery banks.

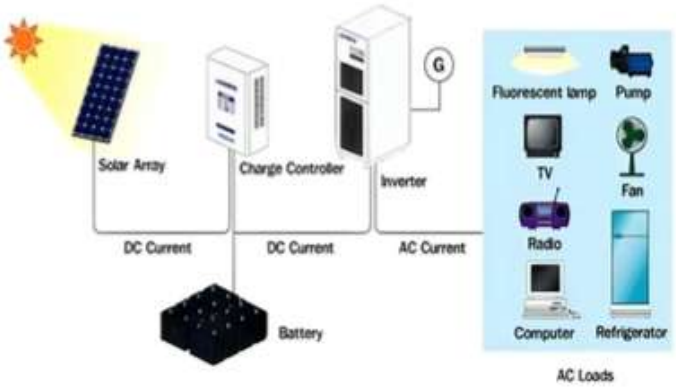


Off-Grid Systems

Connected independent of the utility grid.



DC Off-grid Solar PV System



Standalone Solar PV System

DC Off-grid Solar PV system

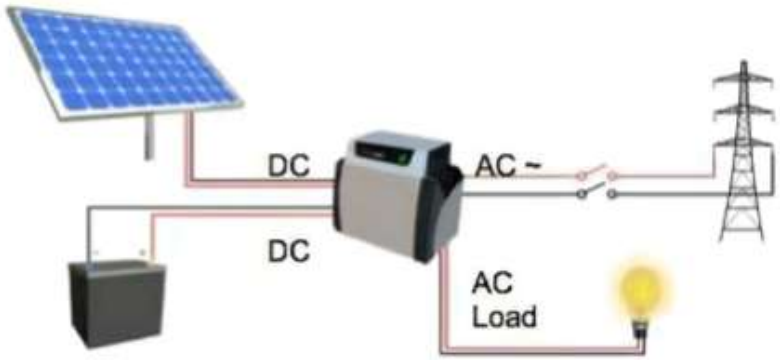
- Uses solar panels to generate DC (or direct current) electricity, powering DC loads.
- Works without an inverter.
- Includes batteries for energy storage, suitable for remote areas without grid access.

Standalone Solar PV system

- Uses solar panels to generate DC electricity, which is then converted to AC electricity using an Inverter. AC is compatible with most household appliances.
- Includes batteries and sometimes backup generators for energy storage, to meet specific energy needs of remote applications like cabins or telecommunications towers.

Hybrid Solar PV Systems

- Hybrid systems combine elements of both grid-connected and off-grid systems, offering greater flexibility and reliability.
- Hybrid system typically include battery storage but remain connected to the grid, allowing users to draw electricity from both sources as needed.



3.5: Technical Parameters and performance of a Solar PV Panel

3.5.1: Technical Parameters and Performance of Solar PV Panel

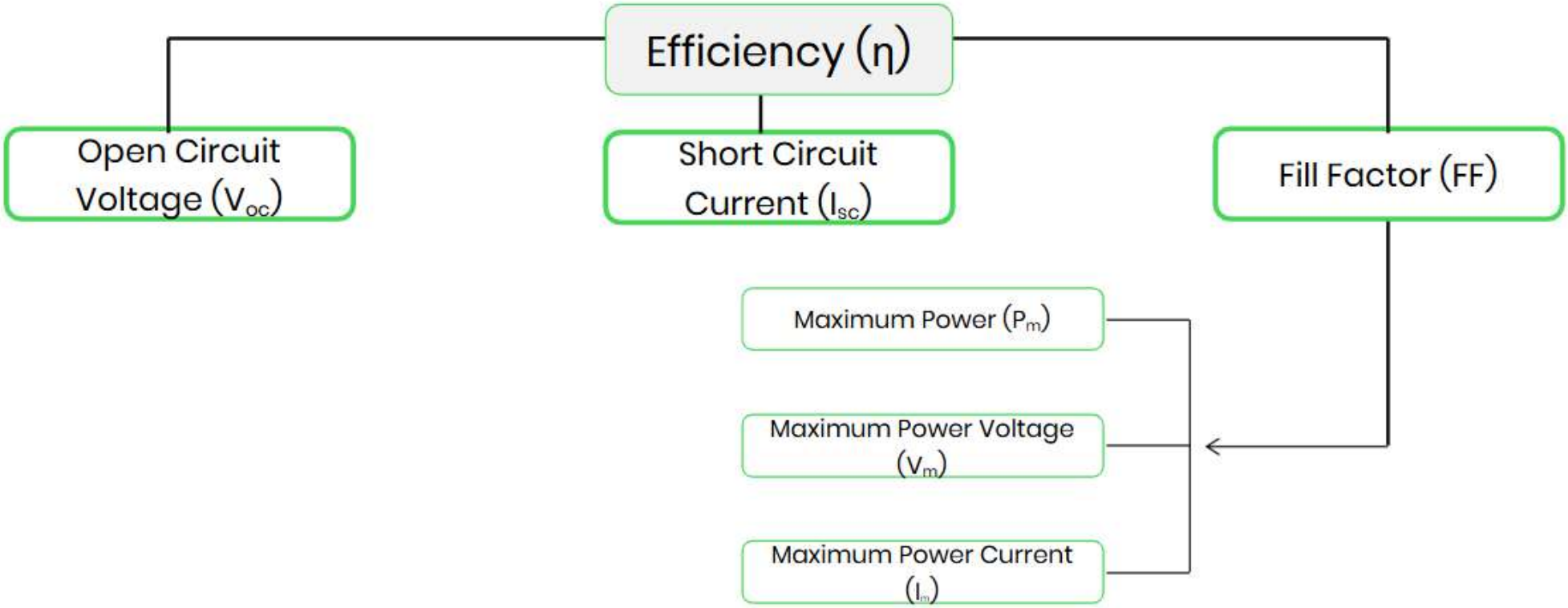
Why are these important?

- For selecting the right solar panel
- For optimizing system design.
- For maximizing energy production.
- For ensuring long-term reliability and performance.








Efficiency (η)
Open Circuit Voltage (V_{oc})
Short Circuit Current (I_{sc})
Fill Factor (FF)
Maximum Power (P_m)
Maximum Power Voltage (V_m)
Maximum Power Current (I_m)

Efficiency (η):

Refers to the percentage of sunlight converted into electricity by the solar panel. Solar PV modules typically exhibit efficiencies ranging from 15-18% under standard testing conditions (STC), with parameters such as irradiance, air temperature, and air mass carefully controlled.

However, their actual performance, as indicated by Nominal Operating Cell Temperature (NOCT), varies due to factors like insolation levels and module temperature, impacting power output in real-world conditions.

Higher efficiency panels produce more electricity for a given area of sunlight.

It is measured by Open Circuit Voltage, Short Circuit Current, Fill Factor (FF).

Open Circuit Voltage (V_{oc}):

The maximum voltage that a solar panel can produce when not connected to a load.

It represents the voltage at which the panel generates no current.

Short Circuit Current (I_{sc}):

The maximum current that a solar panel can produce when its terminals are shorted together.

It represents the current at which the panel generates no voltage.

Fill Factor (FF):

A measure of how effectively a solar cell converts sunlight into electricity.

It is measured by the "squareness" of the IV curve and the ratio of maximum power to the product of open circuit voltage and short circuit current.

An I-V curve represents all of a solar panel's possible operating points at a given cell temperature and light intensity.

Higher fill factors indicate better performance.

To measure the Fill Factor, it is important to consider Maximum Power, Voltage and Current.

Maximum Power (P_m):

The maximum amount of power that a solar panel can produce under standard test conditions.

It is the product of maximum power voltage and maximum power current.

Maximum Power Voltage (V_m):

The voltage at which the solar panel produces its maximum power output.

It is the voltage at which the product of voltage and current is maximized.

Maximum Power Current (I_m):

The current at which the solar panel produces its maximum power output.

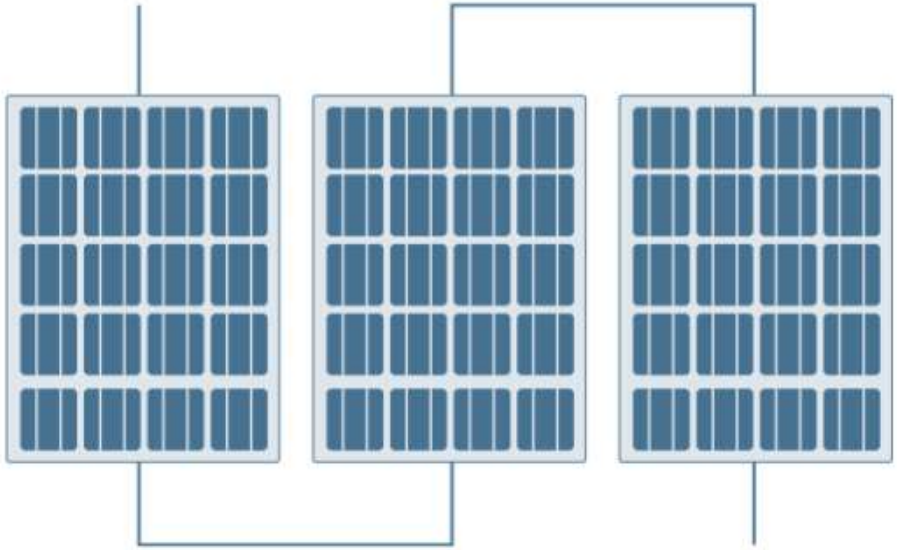
It is the current at which the product of voltage and current is maximized.

Sample values of Technical Parameters of a Solar PV panel:

Parameter	Value
Max Power Pmax (W)	100 W
Power Tolerance (+/-)	0.05%
Max Power Voltage Vmp (V)	18 V
Max Power Current Imp (A)	5.56 A
Open Circuit Voltage Voc (V)	22.3 V
Short Circuit Current Isc (A)	6.1 A
Max. System Voltage Vdc	1000/600

3.5.2: Connect Solar PV modules in Series

String Modules in series



Introduction

Linking multiple panels together in a single electrical circuit where the positive terminal of one module is connected to the negative terminal of the next, creating a continuous flow of electricity.

Voltage and current of modules connected in series



Voltage Increase

When solar PV modules are connected in series, the voltage of each module adds up, resulting in a higher total voltage output for the system.
This is crucial for meeting the voltage requirements of inverters and efficiently transmitting power over longer distances.



Current Consistency

In a series connection, the current flowing through each module remains consistent, as it's determined by the current output of the individual modules.

However, it's important to note that the total current output of the system remains equal to the current output of a single module, as current is the same in a series circuit.



Efficiency Considerations

While series connection increases voltage, it doesn't increase the overall power (wattage) output of the system.

Therefore, balancing the number of modules in series with other factors like shading, temperature effects, and inverter compatibility is important for maximizing system efficiency and performance.

